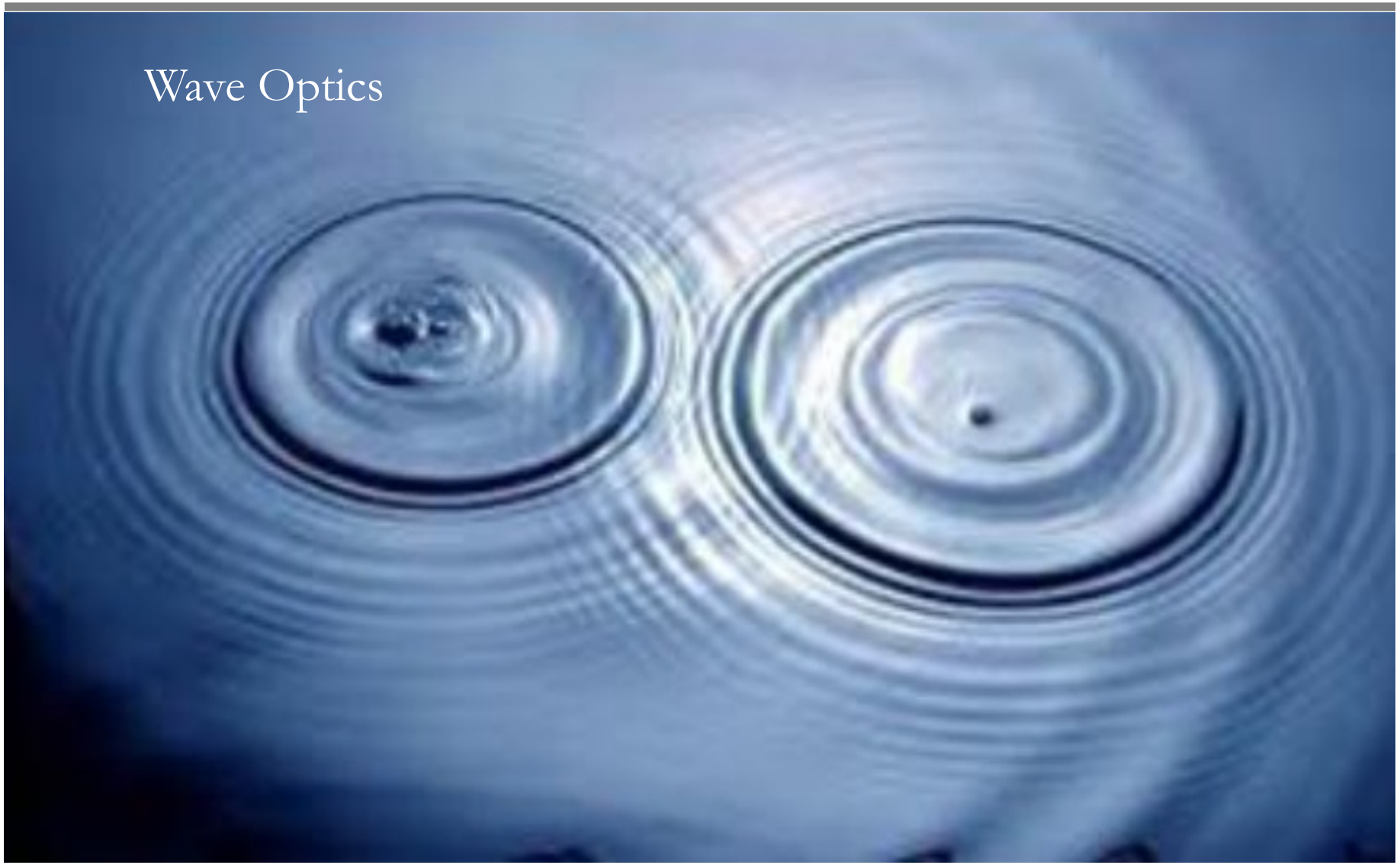

Wave Optics

Interference (4)

4) Two spherical monochromatic light sources, in-phase with same wavelength.

For spherical waves,

$$U(r) = \frac{A}{r} e^{-jkr}$$

Phase difference is related to

$$r_2 - r_1$$

Constructive interference for

$$r_2 - r_1 = n\lambda \qquad n = 0, \pm 1, \pm 2, \ldots$$

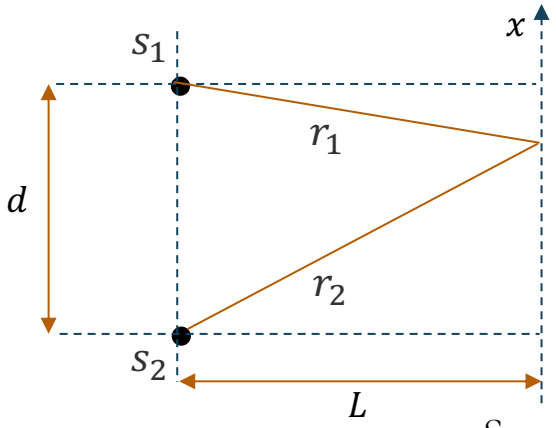
Destructive interference for

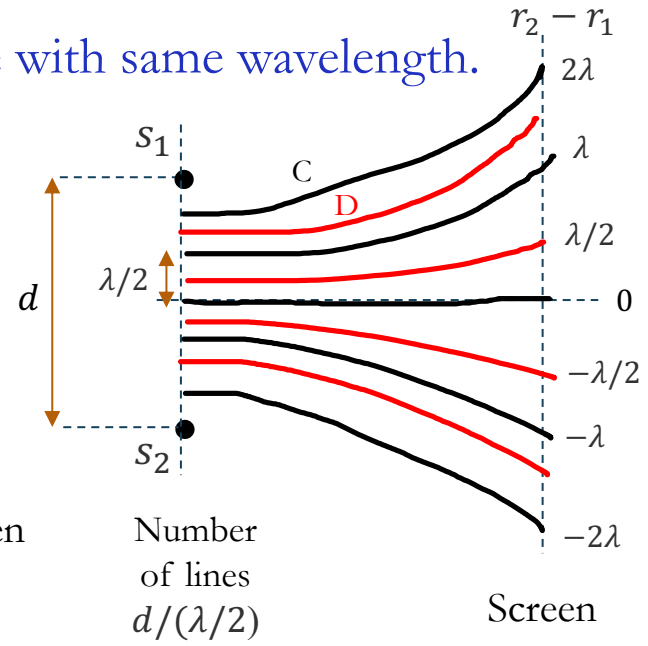
$$r_2 - r_1 = \frac{(2n+1)\lambda}{2}$$

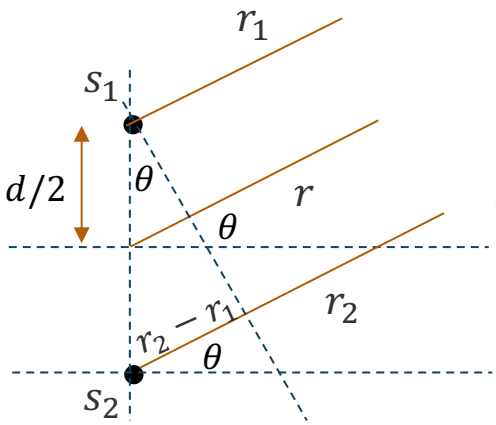
Locations of max/min on the screen (assume $r \gg d$)

Phase difference is given by

$$\delta = \frac{2\pi}{\lambda}(r_2 - r_1) = \frac{2\pi}{\lambda}(d \sin\theta)$$







Constructive interference

$$d \sin\theta = n\lambda \quad \rightarrow \sin\theta_n = n\lambda/d$$

$$x_n = L \tan\theta_n \approx L \sin\theta_n = nL\lambda/d$$

Destructive interference

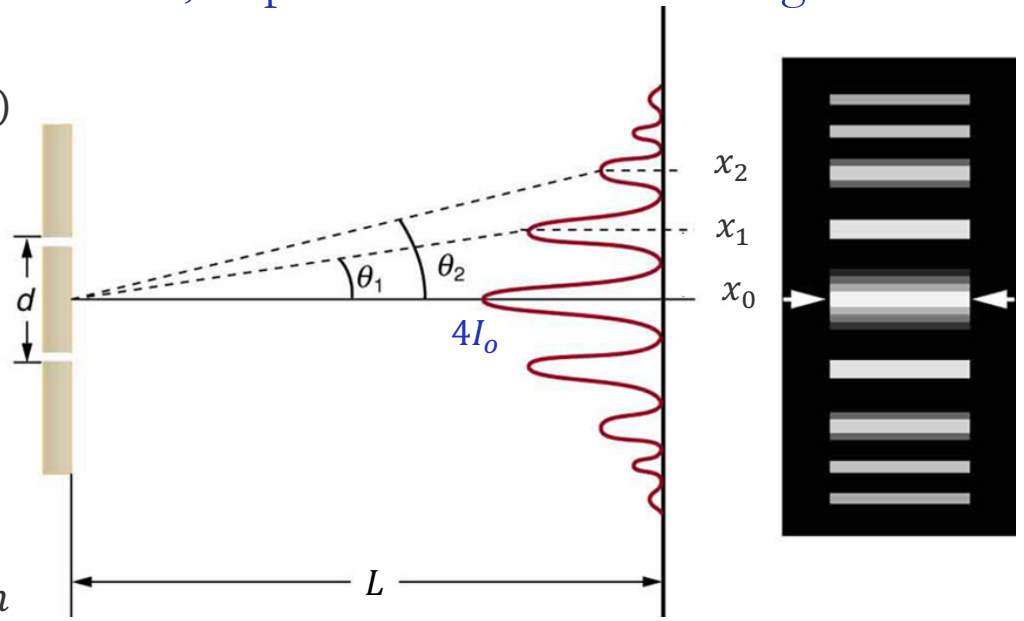
$$x_n = \frac{(2n+1)L\lambda}{2d}$$

Interference (4)

4) Two spherical monochromatic light sources, in-phase with same wavelength.

$$I_n = I_o \left(\frac{\sin \delta_n}{\sin \delta_n/2}\right)^2 \qquad \delta_n = \frac{2\pi}{\lambda}(d \sin \theta_n)$$

$$I_n = 4I_o(\cos \delta_n/2)^2$$



Example.

d=68 μm

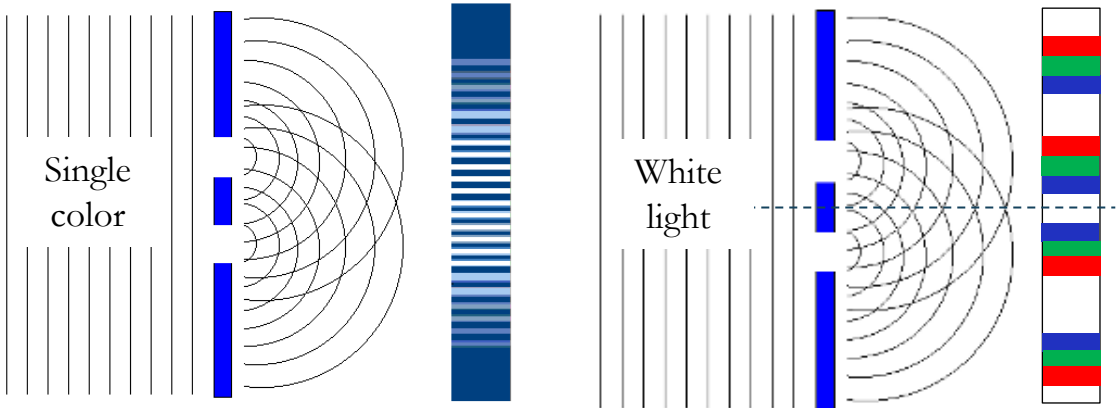
λ=632.8 nm (red)

L=10 m

$\theta_o = 0,\ x_o = 0,\ \theta_1 = 0.412°,\ x_1 = 7.2\ cm$

Question. How are the two sources realized in practice?

Question. If white light is used, how does the interference pattern look like?



Interference (5)

5) Many (N) spherical monochromatic light sources, in-phase with same wavelength. The Gratings problem.

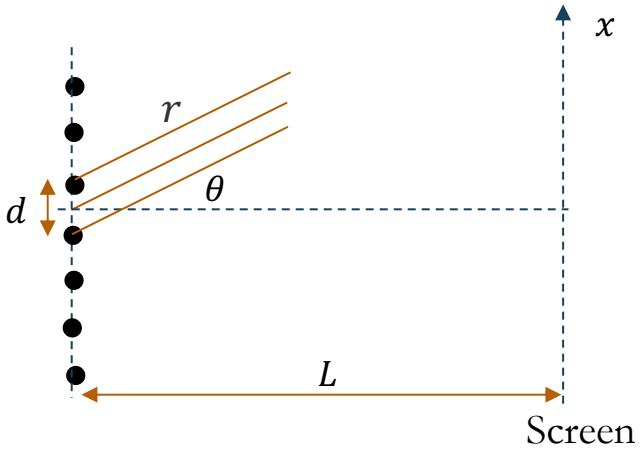
Look at the problem 2 sources at a time.

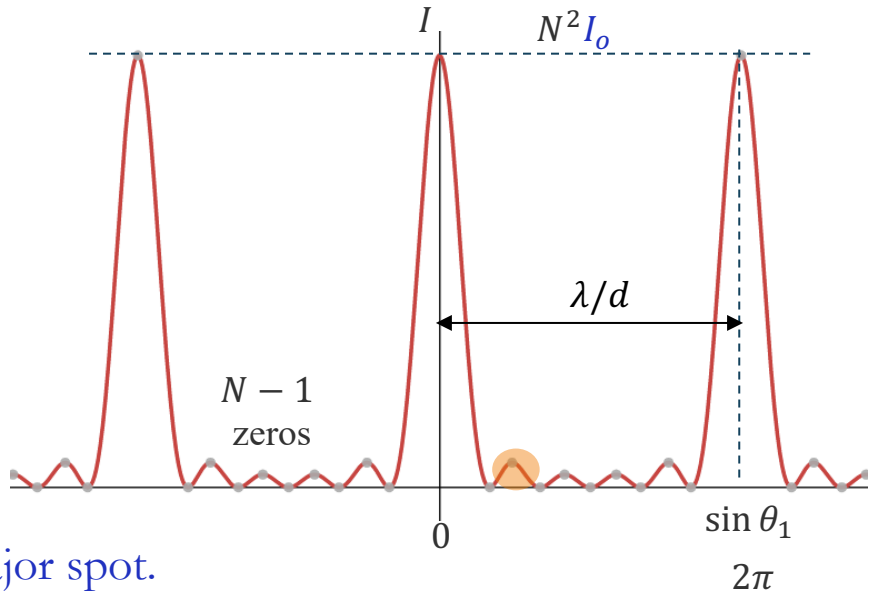
$$\delta = \frac{2\pi}{\lambda}(d \sin\theta_n) = n2\pi$$ Constructive

$$\sin\theta_n = n\lambda/d \qquad x_n \approx nL\lambda/d$$

$$I = I_o\left[\frac{\sin(N\,\delta/2)}{\sin(\delta/2)}\right]^2 \qquad I_{max} = N^2 I_o$$

$$\delta = 0, 2\pi, 4\pi, \ldots$$





Example.
d=2 μm
λ=532 nm (green)
Spot size=2 mm
(1000 lines)
L=5 m → Δx=1.3 m



Find the intensity of the first max after the major spot.

Interference (6)

6) Single slit interference. The Diffraction problem.

Phase difference is related to

$$\beta = \frac{2\pi}{\lambda}(r_2 - r_1) = \frac{2\pi}{\lambda}\frac{a}{2}\sin\theta$$

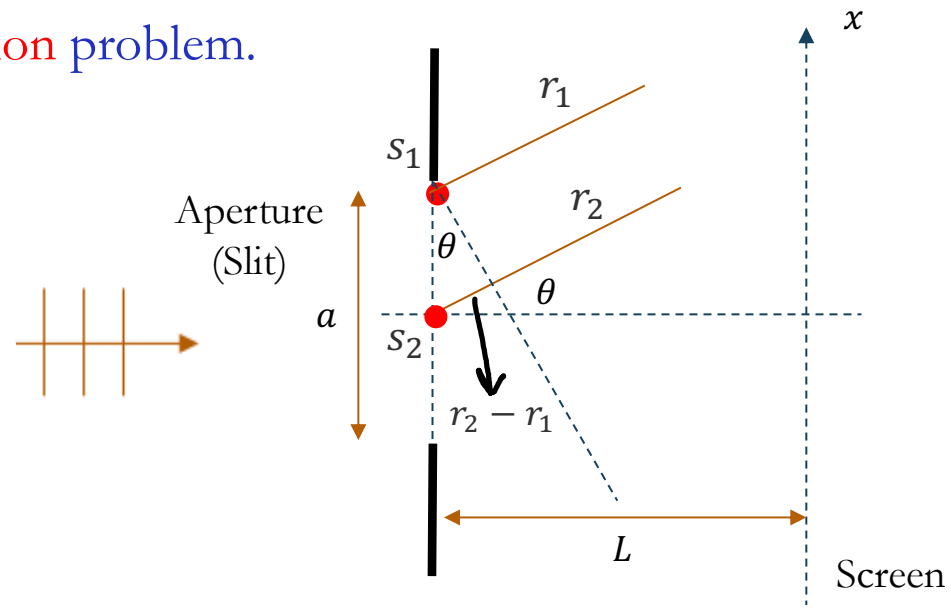
Destructive interference with

$$\frac{a}{2}\sin\theta = n\frac{\lambda}{2} \quad \rightarrow \sin\theta = n\frac{\lambda}{a}$$

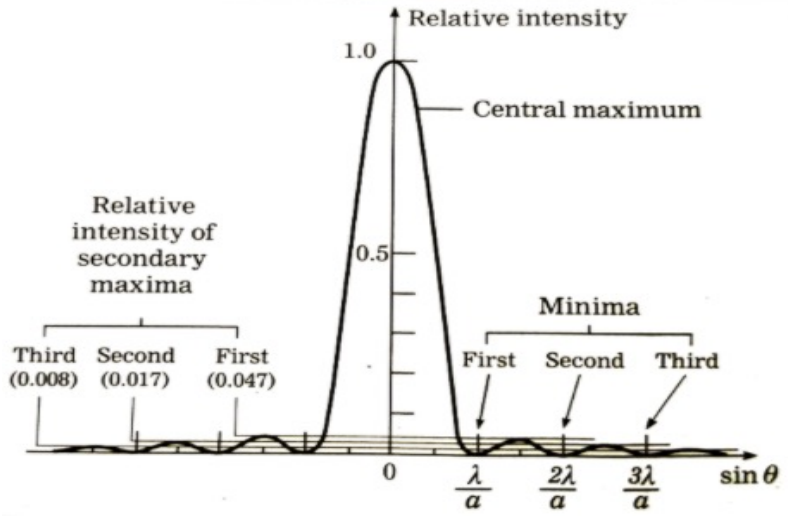
Constructive interference max at $\theta = 0$

Each two points produce the same effect.

Intensity is given by $$I(\beta) \approx I_o\left(\frac{\sin\beta}{\beta}\right)^2$$



$\sin\theta$	β	$\sin\beta$	I	x
0	0	0	I_o	0
λ/a	π	0	o	$\lambda L/a$
$2\lambda/a$	2π	0	o	$2\lambda L/a$



Interference (6)

6) Single slit interference. The Diffraction problem.

Example.1. Find the spot size projected on a screen (L=3 m) by a red laser (630 nm) through an aperture of 0.1 mm.

$$x_1 = \frac{\lambda L}{a} = 2\ cm, x_{-1} = -2\ cm$$ Spot size on screen= 4 cm

Example.2. A 550-nm light through an aperture produces the dark bright spot at 45°. Find the aperture size.

$$a \sin\theta_2 = 2\lambda \qquad a = \frac{2(550\ n)}{\sin 45} = 1.56\ \mu m \qquad \theta_1 = \sin^{-1}\left(\frac{\lambda}{a}\right) = 20.7°$$

Example.3. A 589-nm light goes through a 7.5-μm aperture find the first few angles of the dark spots.

$$\theta_1 = \sin^{-1}\left(\frac{\lambda}{a}\right) = 4.5° \qquad \theta_2 = 9.05° \qquad \theta_{13} = NA$$

Example.4. A 550-nm light goes through a 1-μm aperture find the first few angles of the dark spots.

$$\theta_1 = \sin^{-1}\left(\frac{\lambda}{a}\right) = 33.4° \qquad \theta_2 = NA$$

Interference (6)

6) Single slit interference. The Diffraction problem.

The light intensity of a grating becomes

Some bright spots in the gratings pattern can possibly be nulled by dark spots in the diffraction pattern.

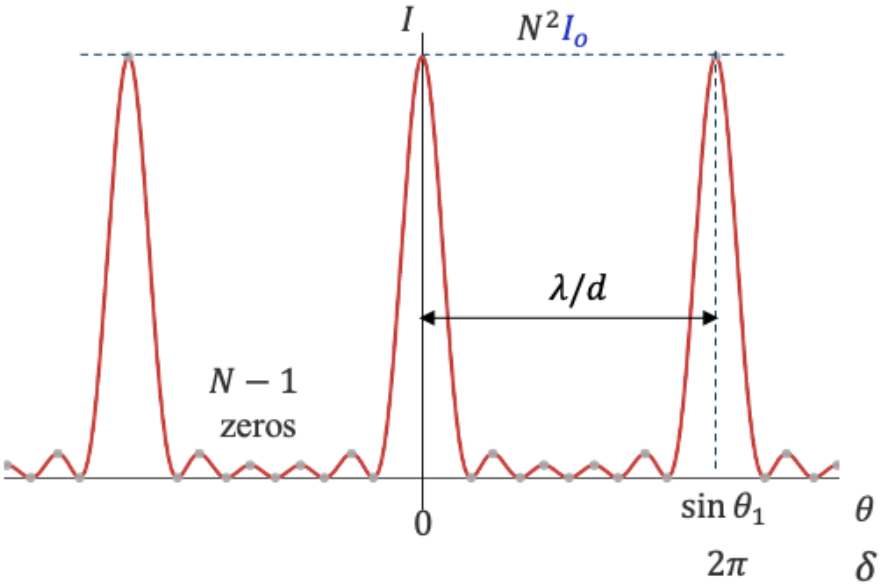
$$I = I_o \left[\frac{\sin(N\,\delta/2)}{\sin(\delta/2)}\right]^2 \cdot \left(\frac{\sin\beta}{\beta}\right)^2$$

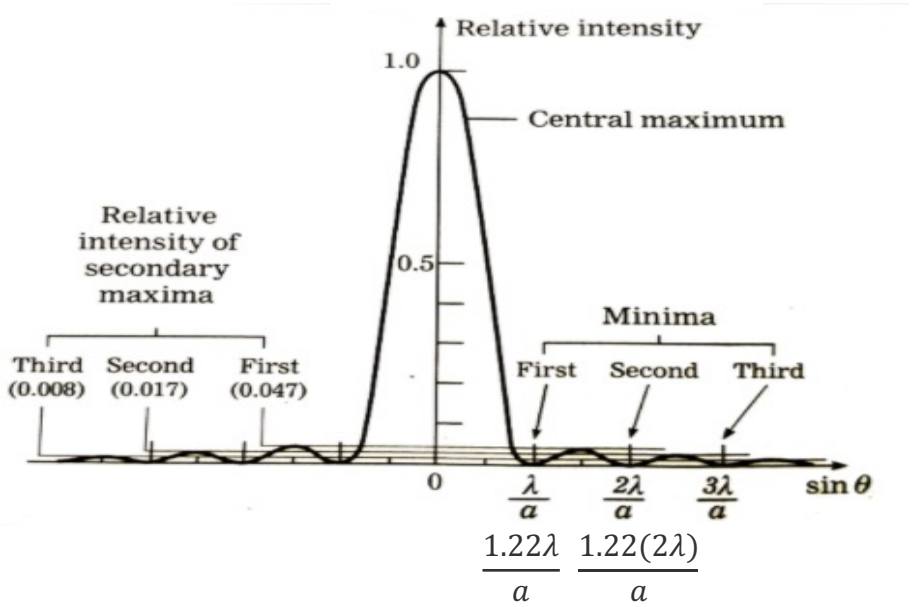
Interference Diffraction

$$\delta = \frac{2\pi}{\lambda}(d\sin\theta)$$

$$\beta = \frac{\pi a}{\lambda}(\sin\theta)$$

For example, if $a = d/5$, the 5th order bright spot in the grating pattern is lost because of the null in the diffraction pattern.

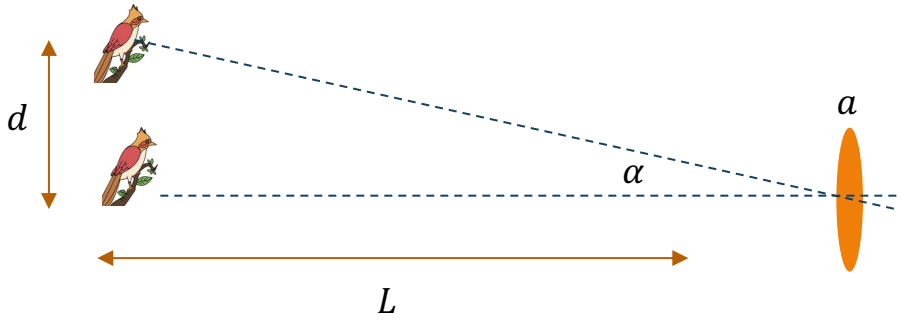


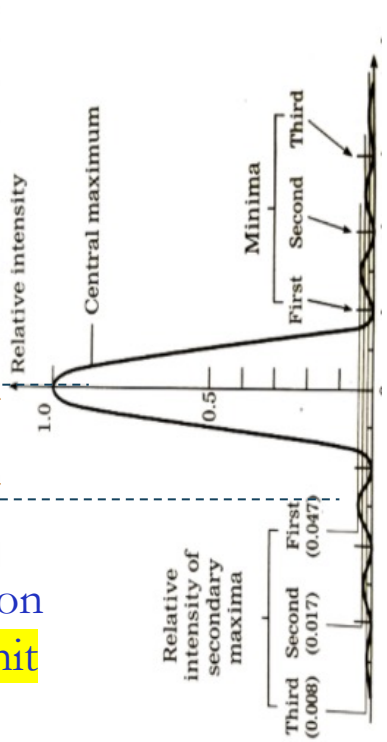


Interference (6)

6) Single slit interference. The Diffraction problem.

For a circular aperture (e.g., human eye) $\frac{\lambda L}{a} \rightarrow 1.22\,\frac{\lambda L}{a}$





$\tan\alpha = \frac{d}{L} \rightarrow \alpha = \tan^{-1}\frac{d}{L}$ $\quad \alpha \geq \Delta\theta$

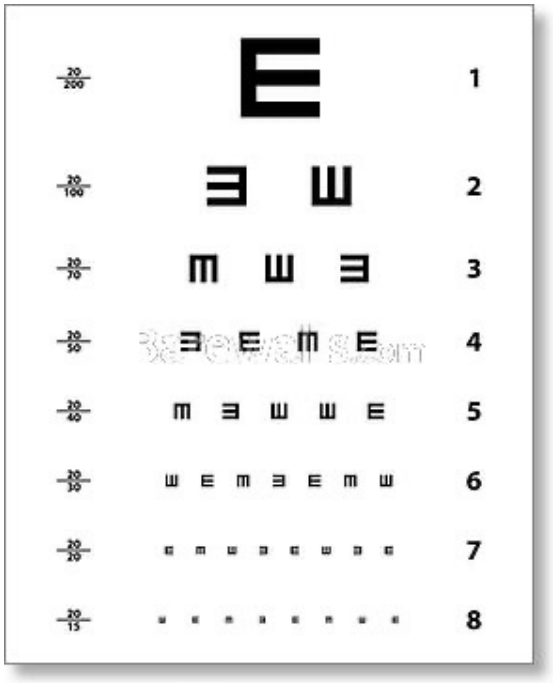
Resolution condition

The diffraction limit

For the human eye, a=4 mm, λ=500 nm $\quad \Delta\theta = 1.5\times10^{-4}$ rad

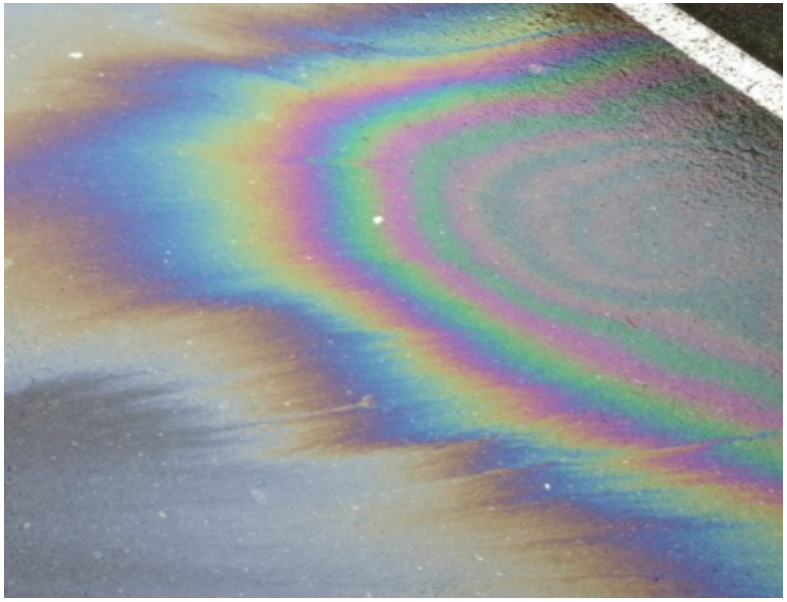
For Hubble telescope, a=2.4 m, λ=500 nm, $\Delta\theta = 0.28\times10^{-6}$ rad

Interference









Electromagnetic Wave Optics

Part I: Mathematical Background and Basic Definitions

The following subjects will be covered

- Maxwell's Equations: Review
- The EM Wave Equation and Solutions.
- The Refractive Index Relation.
- Light Wave Power Density.

EM Wave Optics



EM Wave Optics



EM Wave Optics



EM Wave Optics





EM Wave Optics



Hale-Bopp Comet



Maxwell's Equations

Review of the fundamental laws of electricity and magnetism.

Fields	$\vec{E} = \frac{\vec{F}_e}{q}$	$(\vec{v} \times \vec{B}) = \frac{\vec{F}_m}{q}$
Flux densities	$\vec{D} = \varepsilon \vec{E}$	$\vec{B} = \mu \vec{H}$
Gauss's Law	$\vec{\nabla} \cdot \vec{D} = \rho_v$	$\oint \vec{D} \cdot \vec{ds} = Q_{enc}$
Gauss's Law	$\nabla \cdot \vec{B} = 0$	$\oint \vec{B} \cdot \vec{ds} = 0$
Ampere's Law	$\vec{\nabla} \times \vec{H} = \vec{J}$	$\oint \vec{H} \cdot \vec{dl} = I_{enc}$
Faraday's Law	$\mathcal{E}mf = -\frac{\partial \varphi}{\partial t}$	$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Charge conservation
The continuity equation

$$\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho_v}{\partial t} = 0$$

$$\vec{J} = \sigma \vec{E}$$

Basic Quantities

(V/m) $\vec{E}, \vec{D}$ (C/m^2)

(A/m) $\vec{H}, \vec{B}$ (Wb/m^2)

(A/m^2) $\vec{J}, \rho_v$ (C/m^3)

Material Properties

ε, μ, σ

(F/m) (H/m) (S/m)

Q & A

Send me an email if you have any questions.